2 **(a)** State Newton's second law of motion.

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.....

..... [1]

(b) At a construction site, sand is allowed to fall vertically from a stationary hopper at a steady rate of 100 g s^{-1} on to a horizontal conveyor belt. The belt is driven by an electric motor and moves at a constant velocity of 5.0 cm s^{-1} as shown in Fig. 2.1.

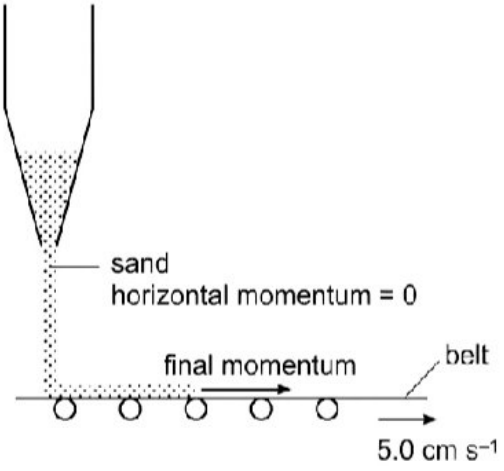


Fig. 2.1.

(i) Using Newton's laws of motion, explain why the sand exerts a horizontal force on the conveyor belt.

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.....

..... [2]

(ii) Show that the magnitude of this horizontal component of force exerted on the sand by the belt is 0.0050 N.

[1]

(iii) Determine

1. the rate at which the motor does work, P, to keep the belt moving at constant velocity, and

P = ... W [1]

2. the rate at which the sand gains kinetic energy, K.

K = ...W [1]

(iv) State the type of collision between the sand and the conveyor belt. With reference to your answers in (b)(iii), describe the energy transfers that occur as the sand is brought to the speed of the belt.

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..... [2]

[Total: 8]